R16

Code No: 131AG

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year I Semester Examinations, May - 2018
ENGINEERING CHEMISTRY
(Common to EEE, ECE, CSE, EIE, IT, ETM)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) What is the role of buffer in the estimation of hardness of water. [2]
b) Explain the cause of Caustic embrittlement in boiler and suggest how to control it. [3]
c) Write applications of fuel cells. [2]
d) Write the Nernst equation and explain the terms. [3]
e) Name the two natural and two synthetic polymers which are used in textile industry. [2]
f) PVC is soft and brittle whereas Bakelite is hard and brittle. Give reason. [3]
g) Why should an ideal fuel have moderate ignition temperature? [2]
h) Gasoline containing tetra ethyl lead is used in internal combustion engines. Give reasons. [3]
i) How RUL test is performed for a refractory? [2]
j) Name the raw materials necessary for the preparation of Portland cement. [3]

PART-B

(50 Marks)

2.a) Explain the steps involved in treatment of sewage water.
b) Explain the principle involved in chlorination and ozonization and write its differences. [5+5]

OR

3.a) With neat diagram discuss Ion exchange process for softening of water.
b) 100ml of a sample of hard water neutralizes exactly 12ml of 0.12N HCl using methyl orange as indicator. What kind of hardness is present? Express the same in terms of an equivalent of $CaCO_3$? [5+5]

4.a) With neat diagram explain the principle, construction and applications of Glass electrode.
b) Differentiate Lithium cell and lithium ion cell. [5+5]

OR

5.a) What is an Electrochemical cell? Explain the construction and reactions of calomel electrode.
b) Explain the construction and functioning of Nickel- Cadmium cell. [5+5]

6.a) What is Co- polymerization? Explain preparation of Nylon 6:6 and Bakelite.
b) Explain the concept of Biodegradable polymers. Discuss two examples and its applications. [5+5]

OR

JNTUH-SEB 14-05-2018 PM

7.a) Differentiate natural rubber and synthetic rubber with suitable examples.
b) With suitable examples differentiate fibre, plastic and elastomer. [5+5]

8.a) What is cracking? Describe the process of moving bed catalytic cracking.
b) The percentage composition of coal is C = 89%, O = 4%, N = 0.5%, S = 1%. Calculate HCV and percentage of hydrogen of the coal sample. [5+5]

OR

9.a) With neat diagram explain refining of petroleum.
b) What is coal? Explain the proximate method for the analysis of coal. [5+5]

10.a) What are composite? Explain classification and applications of compost materials.
b) What is lubricant? Explain viscosity and cloud and pour points of a lubricating oil. [5+5]

OR

11.a) Explain chemical reactions involved in setting and hardening of portland cement.
b) Write short notes on
i) Refractories ii) special cements [5+5]

---ooOoo---

JNTUH USED 14-05-2018PM